# Same job, higher wage? E-3 versus H-1B filings in US Labor Condition Applications, FY2017–2022


| | |
|---|---|
| **Group name** | [Group name] |
| **Dataset** | kaggle.com/datasets/jishnukoliyadan/lca-programs-h1b-h1b1-e3-visa-petitions |
| **Code** | [github.com/… (scripts and README)] |

| # | Name | Roll no. | Institute email | Contribution |
|---|---|---|---|---|
| 1 | [Name 1] | [Roll no.] | [email] | [Contribution] |
| 2 | [Name 2] | [Roll no.] | [email] | [Contribution] |
| 3 | [Name 3] | [Roll no.] | [email] | [Contribution] |
| 4 | [Name 4] | [Roll no.] | [email] | [Contribution] |
| 5 | [Name 5] | [Roll no.] | [email] | [Contribution] |


## 1. Dataset

**Source:** Kaggle, "LCA Programs (H-1B, H-1B1, E-3) Visa Petitions" (CC BY-NC-SA 4.0), an extract of the US Department of Labor's Labor Condition Application (LCA) disclosure data; six yearly CSV files, FY2017–2022, downloaded on 21 October 2026. One row is one LCA: an employer's filing, required before hiring a foreign worker on an H-1B, H-1B1 or E-3 visa, that states the job, the worksite and the *prevailing wage* (the legal minimum DOL sets for that occupation, area and wage level). DOL then certifies or denies it.

**What is raw about it:** the case status is spelled differently before and after FY2020; FY2019's wage column holds the *offered* wage, not the prevailing wage; FY2019's worksite is blank in 88% of rows; wages come in five pay units with typing errors (an "hourly" wage of 46,301, an annual wage of $1 billion); 1,769 spellings for about 1,100 occupation (SOC) titles; 255,125 spellings of employer names; 24% of rows are exact copies of another row.

| | Raw files | After encoding |
|---|---|---|
| Instances (rows) | 3,973,349 | 3,178,679 |
| Attributes (columns) | 11–12 | 210 |
| Instances × attributes | 47,680,188 | 667,522,590 |

**Size check:** the encoded figure is for the 80% exploration set alone (the 20% confirmation set is held out, Section 2): $3{,}178{,}679 \times 210 = 667{,}522{,}590 \geq 3{,}000{,}000$. The 210 features are 197 one-hot levels (the 50 most common SOC titles plus `Other`, worksite and employer state, year, quarter, visa class, status, pay unit, full-time, region) and 13 numeric features and flags. The raw files alone already pass: $3{,}973{,}349 \times 12 \approx 47.7$M.

### Key features

| Feature | Type | Real-world meaning |
|---|---|---|
| `log_pw` | continuous | log of the annual prevailing wage ($15k–$462k after cleaning); response for P1 |
| `case_status` | nominal | DOL decision: certified (90.9%), certified-withdrawn, withdrawn, denied (0.9%); `denied` is the response for P2 |
| `visa_class` | nominal | H-1B (97.6%), E-3 Australian (2.0%), H-1B1 Chile/Singapore (0.4%) |
| `soc_title` | nominal | occupation, 1,123 titles after cleaning; 30% are "Software Developers, Applications" |
| `worksite_state` | nominal | where the worker will work, 55 states and territories |
| `fiscal_year` | discrete | US fiscal year of the decision, 2017–2022 |
| `employer_key` | nominal (ID) | normalised employer name, 165,799 employers; used for grouping, not encoded |
| `employer_volume` | discrete | derived: filings by that employer that year, 1–21,448 |
| `full_time` | nominal | full-time position (98.2%) |

## 2. Data preparation

All rules were fitted on the exploration set and saved; the confirmation set was later cleaned with the same saved rules, never re-fitted. No rows are dropped: every row keeps a valid case decision, so a bad wage is set to missing and flagged instead.

| Issue found | What we did | Why, and not the alternative |
|---|---|---|
| hypotheses would come from EDA | split 80/20 *before* any EDA: stratified group k-fold, grouped by employer, stratified by year × status, seed 42 | testing on the data that suggested a hypothesis is circular. Grouping keeps one employer's near-identical filings out of both sets; a random split would leak them |
| status spelled two ways (upper case before FY2020, title case after) | mapped to 4 labels | otherwise the same decision counts as two levels |

| Issue found | What we did | Why, and not the alternative |
|---|---|---|
| FY2019 wage column is the offered wage | kept as a separate column; FY2019's prevailing wage left missing | 39% of FY2019 values are whole thousands, against under 1% in every other year, and the median per SOC title jumps $+14.5\%$ then falls $-8\%$. Merging would shift FY2019 wages up about 10% |
| 777,688 exact duplicate rows (24%) | **kept**, counted in `n_identical` | there's no case number, and in DOL's source every row is a separate LCA. The duplicate share rises with employer size (18% → 55%) and differs by status (36% certified vs 16% withdrawn), which an export glitch wouldn't cause. Dropping would delete real filings of big employers; their correlation is handled by clustering (Section 5) |
| wrong pay unit ("Hour" $= 46{,}301$) | 427 relabelled to Year | rule: value × unit $> \$1$M/year but the value itself is a normal salary ($15k–$1M); fixing keeps a valid wage |
| five pay units | annualised (×2080, 52, 26, 12) | puts all wages on one scale (2,080 h = DOL's full-time year) |
| 101 wages of $0 or with no unit | set to missing | impossible values; the row's decision is still valid |
| 1,769 SOC title spellings (codes glued on, typos, plurals) | merged to 1,123 titles: letters-only key without plurals, then fuzzy match (similarity $\geq 0.92$) of rare spellings | exact matching leaves e.g. "15-1121.00Computersystems-analysts" as a separate job. We checked the merges by hand; a looser cutoff wrongly joined "Electrical" and "Electronics" engineers |
| wage outliers ($1B "annual" wage) | IQR fence (1.5) on *log* wage *within* SOC title; 21,579 wages set to missing and flagged | log, because wages are right-skewed; within title, because $200k is normal for a surgeon, not a cook. Setting the wage (not the row) to missing keeps the case decision |
| employer country blank (100,916) or "Georgia" | set to USA when the employer state is a US state | the address shows the country; a US state was typed as the country |
| FY2019 worksite 88% blank | flagged, not imputed | worksite and employer state agree in only 52% of rows, and the blank rows differ (93% vs 61% certified), so imputing would invent locations |
| high-cardinality text | one-hot only 10 low-cardinality columns; SOC as top 50 + `Other` (87.7% of rows); employer and job title not one-hot | 132k employers and 375k job titles would add half a million near-empty columns; they enter as filing volume and seniority flags instead |

## 3. What changed after preparation

Exploration set, before and after cleaning (`report/cleaning_log.md`).

| Measure | Before | After |
|---|---|---|
| Rows | 3,178,679 | 3,178,679 (none dropped) |
| Attributes | 17 harmonised | 40 cleaned; 210 after encoding |
| Usable prevailing wages | 2,645,476 | 2,623,796 (21,680 set missing: 21,579 outliers, 101 invalid) |
| Wage range (annual) | $0 – $1B | $15,080 – $462,342 |
| Pay units | 5, 427 mislabelled | 1 (annual) |
| SOC title spellings | 1,769 | 1,123 |
| Job title spellings | 439,495 | 375,194 |
| Employer name spellings | 204,209 | 132,485 employers |
| Blank employer country | 100,916 | 0 |
| Exact duplicate rows | 777,688 | 777,688, kept and counted |
| Missing cells (not by design) | 1.92% | 1.56% |

Cleaning turned the wage into one comparable annual number: the lowest cleaned wage is exactly $\$7.25 \times 2{,}080$, the US federal minimum wage, a sign the unit fixes worked. Its log is almost symmetric (skew $-0.04$, against $1.14$ for dollars). Harmonising occupation and employer names is what makes the within-occupation and within-employer comparisons of Section 5 possible.

## 4. Basic questions and problem statement

**Basic questions**

1. Which employers file the most? (Cognizant, Infosys, TCS, Google, Amazon; the top 10 file 15%.)
2. Which jobs? (Mostly software: computer titles are 65% of filings; Fig. 1.)
3. How many filings are certified? (91%; only 0.87% are denied.)
4. Have wages risen? (Median prevailing wage $+33\%$ from FY2017 to FY2022.)
5. Do some regions pay more? (West $105k vs Midwest $78k.)

**Our problem statements**

**P1 (primary).** For the same occupation, worksite state and fiscal year, do E-3 (Australian) filings carry a different prevailing wage than H-1B filings, and does any difference remain within the same employer?

**P2.** Are E-3 filings denied more or less often than H-1B filings from employers of the same size in the same year, and within the same employer?

**How they are novel:** the basic questions summarise one variable at a time. The prevailing wage is set by DOL from occupation, area and wage level; the visa isn't an input, so "no difference" is a natural default for P1. A difference would mean employers place the two groups at different wage levels, which the data records only indirectly (Fig. 4). P1 holds fixed what the basic questions show moves wages most (occupation, place, year) and then the employer. P2 asks whether a 5.8× raw denial gap survives the main confounder found in EDA, employer size (Fig. 3), a possible Simpson's paradox.

## 5. Statistical inferences

**Rows tested:** the held-out confirmation set: 794,670 filings from 33,314 employers that never appear in the exploration set, cleaned with the saved rules. E-3 vs H-1B filings only (H-1B1 is 0.4% and a separate program). Wage tests use FY2017–18 and FY2020–22 (FY2019 has no prevailing wage) and rows with a cleaned wage and known worksite.

**Common design, fixed before the data was opened:** each test is the coefficient $\beta$ in

$$y_i = \beta\,\mathrm{E3}_i + \alpha_{c(i)} + \varepsilon_i,$$

where $\alpha_c$ is a fixed effect for the comparison cell (e.g. SOC × state × year). Fixed effects compare E-3 and H-1B filings only inside the same cell; they are removed by demeaning within cells. Standard errors are **clustered by employer** (CR1) with a $t$ reference on $G-1$ df ($G$ = employers): one employer's filings are near-copies, so they are not independent. Four tests, two-sided, family-wise $\alpha = 0.05$ with **Holm** correction. With this many rows almost anything is significant, so each result is also judged by whether its 95% CI clears the smallest meaningful difference (MID): *meaningful* (CI beyond the MID), *detectable, possibly too small* (CI overlaps it), *negligible* (inside it, confirmed by TOST) or *inconclusive*. Power at the MID, from exploration-set SEs: T1 1.00, T2 0.999, T3 0.16, T4 0.93.

**Test 1 (P1): do E-3 filings carry a different prevailing wage for the same job, state and year?**

**a) Hypotheses, written before running the test**

| | |
|---|---|
| $H_0$ | $\beta = 0$ |
| $H_1$ | $\beta \neq 0$ |
| **Significance level** | $\alpha = 0.05$ family-wise (Holm, 4 tests) |
| **Side** | two-sided |
| **Test** | $t$-test on $\beta$, fixed effects for SOC × state × year, employer-clustered SE |
| **Smallest difference that matters** | 5% ($\lvert\beta\rvert \geq \log 1.05 = 0.049$) |

$y$ = log annual prevailing wage, so $\beta$ is the E-3 minus H-1B difference in log points (≈%). Two-sided, because choosing a direction after EDA would be data snooping. 5%, because it is about $4,400 at the median wage, about one year of wage growth here, and half the gap between "senior" and other titles in the same job: a smaller gap wouldn't place a worker at a different wage level.

**b) Other tests we considered**

Two groups and a continuous, symmetric response suggest a two-sample $t$-test, but the groups must be compared within the same occupation, state and year (Figs. 1, 2), and filings aren't independent. A fixed-effects regression does both comparisons at once for 2,729 cells, and clustering by employer handles the dependence.

| Also considered | Why we did not use it |
|---|---|
| Welch $t$-test on relative wage | treats filings as independent; its SE (0.0027) is 2.4× too small. Run as a check: $t = 20.7$, same decision |
| Mann–Whitney U | tests a shift in ranks, which can't be compared with a 5% MID; also assumes independence. Run as a check: $p < 10^{-100}$ |
| Mixed model, employer random effects | assumes employer effects are unrelated to visa class, the very thing Test 2 questions |
| Raw two-group comparison | mixes job, state and year composition into the gap |

**c) Result**

| | |
|---|---|
| **Rows used** | 408,671 (10,433 E-3), 19,299 employers |
| **Estimate** | $+9.1\%$ ($\hat{\beta} = 0.087$; raw gap $+9.6\%$) |
| **95% confidence interval** | $+7.7\%$ to $+10.5\%$ |
| **Test statistic (d.f.)** | $t = 13.66$ (19,298) |
| **p-value (Holm)** | $3.0 \times 10^{-42}$ ($8.9 \times 10^{-42}$) |
| **Wild-cluster bootstrap p** | $\leq 0.0005$ (1,999 draws) |
| **Decision** | reject $H_0$ |
| **Verdict** | meaningful: the whole CI is above 5% |

For the same occupation, state and year, E-3 filings carry a prevailing wage about 9% higher than H-1B filings, well past the 5% that matters. Occupation, state and year explain almost none of the raw $+9.6\%$ gap. The estimate is stable

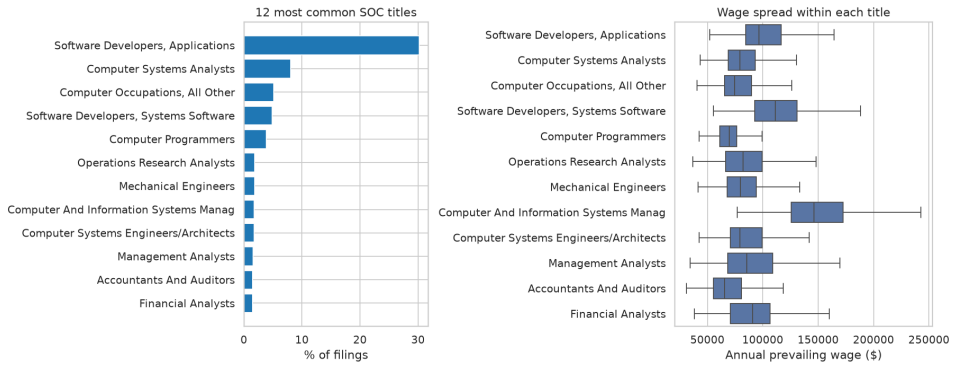



**Figure 1:** This is mostly a computer-jobs dataset: "Software Developers, Applications" alone is 30% of filings, and within a title the 75th-percentile wage is 1.4× the 25th, so occupation has to be held fixed before comparing anything.

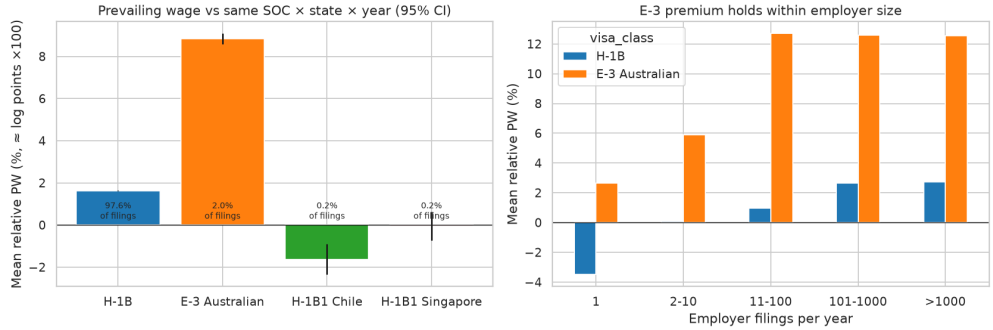



**Figure 2:** Relative prevailing wage = log wage minus the median of the same SOC title × state × year. E-3 filings sit $+8.8\%$ above comparable jobs on average, H-1B $+1.6\%$ (left); the gap holds at every employer size above 10 filings a year (about $+13\%$ vs $+1$ to $+3\%$, right).

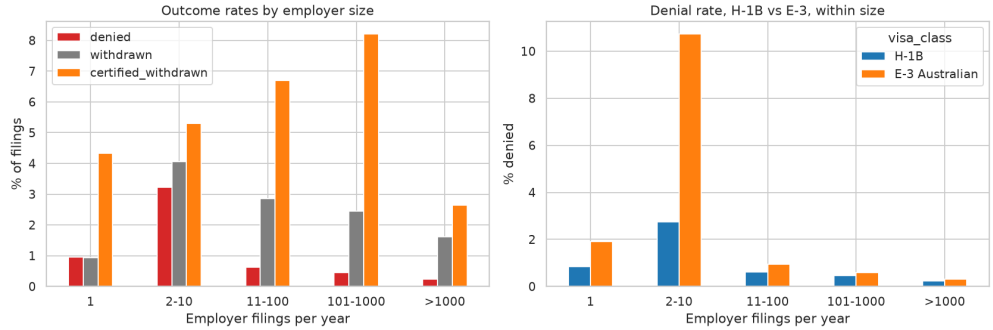



**Figure 3:** Denial rates peak at employers with 2–10 filings a year (3.2%), against 0.23% for the largest filers (left). Half of E-3 filings come from employers with $\leq$10 filings, against 18% of H-1B, yet E-3 is denied more even within the same size group (right).

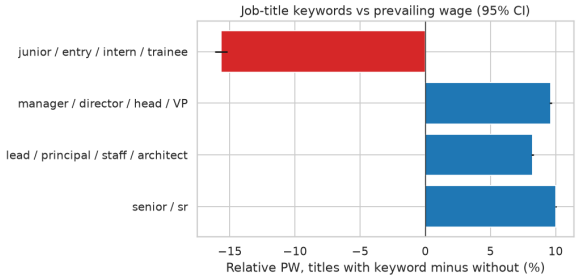



**Figure 4:** Job-title keywords move the relative prevailing wage as expected (senior $+10\%$, junior $-16\%$), so relative wage tracks the wage level an employer chose, a column this extract doesn't have.

across years ($+8$ to $+12\%$, Fig. 5) and unchanged without the largest employer.

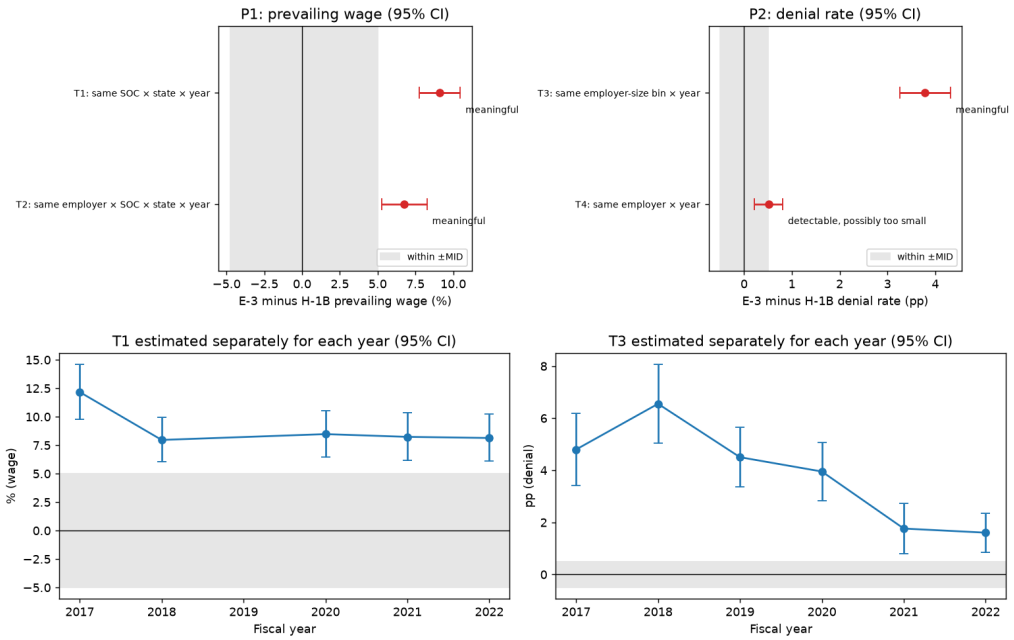



**Figure 5:** Top: all four estimates with 95% CIs; the grey band is ±MID. Tests 1–3 clear it; Test 4 straddles it. Bottom: Tests 1 and 3 estimated year by year (exploratory, not pre-registered). The wage gap is stable at $+8$ to $+12\%$; the denial gap shrinks after FY2020.

### Test 2 (P1): does the wage gap remain within the same employer?

#### a) Hypotheses, written before running the test

| | |
|---|---|
| $H_0$ | $\beta = 0$ |
| $H_1$ | $\beta \neq 0$ |
| **Significance level** | $\alpha = 0.05$ family-wise (Holm) |
| **Side** | two-sided |
| **Test** | as Test 1, cells = employer × SOC × state × year |
| **Smallest difference that matters** | 5% |

Only employers that file both visas for the same job, state and year contribute. If the Test 1 gap vanished here, it would only mean that higher-paying employers hire Australians; if it remains, the same employer files the two groups differently.

#### b) Other tests we considered

Holding the employer fixed removes its pay policy, industry and lawyer, the confounders Test 1 leaves open; fixed effects do this without assuming anything about how employers differ.

| Also considered | Why we did not use it |
|---|---|
| Paired $t$-test on cell means | weights a cell with 1 E-3 and 500 H-1B filings like a balanced one |
| Matching E-3 to H-1B filings | discards most H-1B filings for no gain in bias |
| Mixed model | random effects assume away the employer confounding being tested |

#### c) Result

| | |
|---|---|
| **Rows used** | 71,060 (3,356 E-3), 695 employers, 1,906 cells |
| **Estimate** | $+6.8\%$ ($\hat{\beta} = 0.065$) |
| **95% confidence interval** | $+5.3\%$ to $+8.3\%$ |
| **Test statistic (d.f.)** | $t = 9.24$ (694) |
| **p-value (Holm)** | $2.9 \times 10^{-19}$ ($5.8 \times 10^{-19}$) |
| **Wild-cluster bootstrap p** | $\leq 0.0005$ |
| **Decision** | reject $H_0$ |
| **Verdict** | meaningful: the whole CI is above 5% |

The same employer files its E-3 workers at a prevailing wage about 7% higher than its H-1B workers in the same job, state and year. About 75% of the Test 1 gap remains (comparing two populations, as Test 2 covers only employers filing both visas), so the premium is mostly *how* employers file, not *which* employers hire Australians.

**Test 3 (P2): are E-3 filings denied more often by employers of the same size?**

**a) Hypotheses, written before running the test**

| | |
|---|---|
| $H_0$ | $\beta = 0$ |
| $H_1$ | $\beta \neq 0$ |
| **Significance level** | $\alpha = 0.05$ family-wise (Holm) |
| **Side** | two-sided |
| **Test** | linear probability model, fixed effects for employer-size group × year, employer-clustered SE |
| **Smallest difference that matters** | 0.5 percentage points |

$y$ = 1 if denied. Size groups: 1, 2–10, 11–100, 101–1000, >1000 filings a year; all six years are used. 0.5 pp, because the H-1B denial rate is 0.78%, so it means a $\geq$1.6× higher risk, and it is over half the whole FY2017–22 fall in the denial rate (1.36% → 0.49%).

**b) Other tests we considered**

Employer size is the main confounder (Fig. 3), so the comparison is within size group and year. A linear probability model gives the gap directly in percentage points, the unit of the MID, and its clustered SE needs no functional form.

| Also considered | Why we did not use it |
|---|---|
| Cochran–Mantel–Haenszel test | the classic stratified 2×2 test, but assumes independent filings. Run as a check: pooled OR 3.82, $p < 10^{-100}$ |
| Logistic regression | gives an odds ratio, not pp. Run as a check, same fixed effects and clustering: OR 3.80, $z = 24.9$ |
| $\chi^2$ test on the raw 2×2 table | ignores the size confounder the question is about |

**c) Result**

| | |
|---|---|
| **Rows used** | 791,227 (14,587 E-3), 32,956 employers |
| **Estimate** | $+3.79$ pp (E-3 5.32%, H-1B 0.77% raw; raw gap $+4.55$ pp) |
| **95% confidence interval** | $+3.26$ to $+4.31$ pp |
| **Test statistic (d.f.)** | $t = 14.12$ (32,955) |
| **p-value (Holm)** | $4.0 \times 10^{-45}$ ($1.6 \times 10^{-44}$) |
| **Wild-cluster bootstrap p** | $\leq 0.0005$ |
| **Decision** | reject $H_0$ |
| **Verdict** | meaningful: the whole CI is above 0.5 pp |

Among employers of the same size in the same year, E-3 filings are denied 3.8 pp more often; employer size explains only 17% of the raw gap. The power check expected only 0.16 power at the MID, but the effect turned out about eight times larger. *Exploratory, not pre-registered:* the gap shrinks from $+6.6$ pp in FY2018 to $+1.6$ pp in FY2022 (Fig. 5).

**Test 4 (P2): does the denial gap remain within the same employer?**

**a) Hypotheses, written before running the test**

| | |
|---|---|
| $H_0$ | $\beta = 0$ |
| $H_1$ | $\beta \neq 0$ |
| **Significance level** | $\alpha = 0.05$ family-wise (Holm) |
| **Side** | two-sided |
| **Test** | as Test 3, cells = employer × year |
| **Smallest difference that matters** | 0.5 percentage points |

Only employers filing both visas in the same year contribute. Holding the employer fixed removes its experience with the process and its form-filling practice, which size groups only approximate.

**b) Other tests we considered**

| Also considered | Why we did not use it |
|---|---|
| Conditional (fixed-effects) logit | planned as a check, but the exact likelihood failed on employer-years with thousands of filings. With one binary regressor its score test equals the CMH test by employer × year, which we ran instead: OR 1.89, $p = 7.6 \times 10^{-6}$ |
| Logit with employer dummies | 1,547 employers' dummies give biased estimates when most employer-years have no denial (incidental-parameter problem) |

**c) Result**

| | |
|---|---|
| **Rows used** | 382,911 (8,003 E-3), 1,547 employers |
| **Estimate** | $+0.51$ pp (H-1B rate at these employers 0.30%) |
| **95% confidence interval** | $+0.22$ to $+0.81$ pp |
| **Test statistic (d.f.)** | $t = 3.44$ (1,546) |
| **p-value (Holm)** | $5.9 \times 10^{-4}$ ($5.9 \times 10^{-4}$) |
| **Wild-cluster bootstrap p** | $\leq 0.0005$ |
| **Decision** | reject $H_0$ |
| **Verdict** | detectable, possibly too small: the CI straddles 0.5 pp (TOST $p = 0.54$) |

Within the same employer the gap is real but small in absolute terms; we can't tell whether it reaches the 0.5 pp that matters. Relative to these employers' low H-1B rate (0.30%) it is about 2.7×. Across the two populations, 86% of the Test 3 gap disappears when the employer is held fixed: most of the E-3 denial gap is *between* employers.

## 6. Limitations

- **Association, not cause.** Australians may simply be hired into more senior roles; worker experience and education aren't in the data. We show that filings differ, not that the visa causes it.
- **Prevailing wage, not wage paid.** It is the legal minimum, and the wage-level column was cut from this extract; relative wage is an indirect proxy (Fig. 4). State is coarser than the metro area DOL actually uses.
- **FY2019 left out of the wage tests**, since its file has no prevailing wage, and most of its worksites are blank.
- **Duplicates kept.** If some identical rows are export errors, big employers are over-weighted; clustering keeps the SEs honest, but we didn't re-run the tests on deduplicated data.
- **Tests 2 and 4 cover only employers filing both visas**, which are larger than average.
- **Denial here is DOL's check of the form**, not the visa decision, which USCIS makes later.
- **Exploratory extras.** The year-by-year results were not pre-registered and are not Holm-corrected.